Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 38)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 The Universal Soil Loss Equation (USLE) is A = R x K x LS x C x P. Table 1 gives factor values for a farm.

Table 1

Factor	Description	Value
R	Rainfall erosivity for the farm	110
K	Soil erodibility: loam	0.67
LS	Slope length and gradient: 120 m, 6%	1.35
C (crop type x tillage)	Beans (0.50) with spring ploughing (0.90); or wheat (0.35) with mulch tillage (0.60)	see question
P	Ploughing up and down slope 1.0; contour ploughing 0.50; strip cropping with contour ploughing 0.25	see question

Table 2

Soil erosion class	Rate of soil loss / t ha^{-1} yr^{-1}
Low	under 5
Moderate	5-10
High	10-20
Very high	20-30
Severe	over 30

0 1 . 1 Calculate the annual soil loss for beans with spring ploughing, ploughed up and down the slope. Give your answer to one decimal place.

[2 marks]

Answer _____ t/ha/yr

0 1 . 2 Calculate the annual soil loss for wheat with mulch tillage and strip cropping with contour ploughing.

[2 marks]

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Answer [ ] t/ha/yr

0 1 . 3 The natural rate of soil formation on the farm is 1 t ha^{-1} yr^{-1}. Use your answers and Table 2 to evaluate the sustainability of the two farming systems.

[3 marks]

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0 1 . 4 Explain why the USLE gives only an estimate of soil loss.

[2 marks]

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0 1 . 5 Suggest which single factor in the USLE a farmer can change most easily.

[1 mark]

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0 2 Table 1 compares the energy used to produce 1 tonne of aluminium from bauxite ore with the energy used to produce 1 tonne from recycled aluminium cans.

Table 1

Source	Energy used / GJ per tonne	CO_2 emitted / tonnes per tonne of aluminium
Bauxite ore (extraction, refining, smelting)	210	12
Recycled cans (collection, sorting, remelting)	12	0.7

0 2 . 1 Calculate the percentage energy saving from recycling aluminium rather than extracting it from ore.

[2 marks]

Answer ..... %

0 2 . 2 Give **three** advantages of recycling metals other than energy saving.

[3 marks]

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0 2 . 3 Explain **four** difficulties that reduce the effectiveness of recycling schemes.

[4 marks]

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0 2 . 4 Explain what is meant by **Cradle to Cradle** design.

[1 mark]

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0 3 The Aral Sea in central Asia was once the fourth largest lake in the world. From the 1960s water from the two rivers feeding it (the Amu Darya and Syr Darya) was diverted to irrigate cotton and rice. Table 1 shows changes in the lake.

Table 1

Year	**1960**	**1980**	**2000**	**2015**
Lake area / km^2	68 000	52 000	24 000	7 000
Salinity / g dm^{-3}	10	18	45	over 100
Commercial fish catch / thousand tonnes	40	10	0	0

0 3 . 1 Calculate the percentage reduction in the area of the Aral Sea between 1960 and 2015.

[2 marks]

Answer ☐ %

0 3 . 2 Explain why the salinity of the lake increased as the area fell.

[2 marks]

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0 3 . 3 Explain why the fish catch fell to zero, and describe **two** other environmental consequences of the shrinking of the lake.

[4 marks]

0 3 . 4 Suggest why disagreements over the rivers continue between the countries of the region.

[2 marks]

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0 4 . 1 Which of the following oil spill treatments is most likely to kill organisms living deep in beach sediments that were not affected by the oil? Tick one box.

[1 mark]

A Absorbent mops ☐

B Steam washing ☐

C Inflatable booms ☐

D Polymerising agents ☐

0 4 . 2 Explain how polymerising agents make an oil spill easier to clean up.

[2 marks]

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0 4 . 3 Explain why deoxygenation is a more common problem in water than in air.

[2 marks]

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0 5 . 1 Explain the difference between a **mutagen**, a **carcinogen** and a **teratogen**.

[3 marks]

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0 5 . 2 Explain why a birth abnormality caused by a teratogen cannot be inherited by future generations, whereas one caused by a mutagen may be.

[2 marks]

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0 5 . 3 Distinguish between **somatic** and **gonadic** effects of a mutagen.

[2 marks]

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0 5 . 4 Explain what is meant by **synergism** and give an example.

[3 marks]

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0 6 . 1 Describe how DNA databases are used to control the illegal trade in tropical timber and ivory.

[3 marks]

0 6 . 2 Explain the difference between radio / GPS tracking and acoustic tracking and state when each is used.

[2 marks]

0 6 . 3 Give **three** types of carrier system used to move sensors to study areas and one advantage of drones for ecological monitoring.

[3 marks]

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0 6 . 4 Explain how motion-sensitive cameras and CCTV reduce disturbance to wildlife compared with direct observation.

[2 marks]

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0 7 . 1 Which of the following water sources is usually the most convenient for large communities? Tick one box.

[1 mark]

A Desalinated seawater ☐

B River water ☐

C Rainwater catchment ☐

D Newly discovered aquifers ☐

0 7 . 2 Explain why the quality requirements for water differ between uses, using **two** named uses.

[2 marks]

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0 7 . 3 Explain why inter-basin transfers may be used and give **one** disadvantage.

[2 marks]

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0 8 Table 1 shows the bypass ratio and first year of use of jet engines used on four airliners.

Table 1

Engine	Aircraft	Bypass ratio	First use
RR Olympus	Concorde	0 : 1	Early 1950s
RR Conway	Boeing 707	0.25 : 1	Late 1950s
P&W JT9D	Boeing 747	5 : 1	1969
RR Trent 1000	Boeing 787	10 : 1	2011

0 8 . 1 Explain how a high bypass ratio reduces engine noise.

[3 marks]

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0 8 . 2 Describe **two** other features of aircraft design that reduce noise.

[2 marks]

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0 8 . 3 Explain how each of the following airport operating procedures reduces noise nuisance.

[4 marks]

Constant descent angle .....

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Night flight restrictions .....

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Noise-based landing charges / quota count .....

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Multiple runways .....

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0 8 . 4 Explain why aircraft are noisier on take-off than on landing.

[1 mark]

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0 9 A nuclear fuel reprocessing plant discharges small amounts of radioactive material in liquid effluent to a coastal sea.

0 9 . 1 Explain why it cannot be assumed that the discharge will be diluted to insignificant levels.

[2 marks]

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0 9 . 2 Give **four** factors that would be included in a Critical Pathway Analysis of this discharge.

[4 marks]

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0 9 . 3 Explain how the results of the analysis reduce the cost of environmental monitoring.

[2 marks]

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0 9 . 4 Explain why Critical Group Monitoring measures exposure rather than damage to health.

[2 marks]

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1 0 Table 1 shows the three forms of oxygen found in the stratosphere.

Table 1

Form of oxygen	Number of atoms per molecule
Monatomic oxygen (O)	1
Diatomic oxygen (O_2)	2
Triatomic oxygen / ozone (O_3)	3

1 0 . 1 Explain how the three forms of oxygen protect life at the surface from ultraviolet radiation.

[3 marks]

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1 0 . 2 Give **two** other ways in which the atmosphere supports life.

[2 marks]

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1 0 . 3 Explain why the atmosphere is thermally stratified into layers.

[1 mark]

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1 0 . 4 The troposphere and stratosphere are the layers most affected by human activities. Discuss how human activities alter the energy processes in these two layers and evaluate the difficulties of predicting the consequences.

[9 marks]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.

[25 marks]

OR

1 1 . 2 Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS